SYNTHETIC INPUT

Acting on these findings is modelled to improve health and reduce healthcare spending.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

INCREMENTAL QALYS

0.082

about 30 days of healthy life, on average, for a person with this pattern of findings

INCREMENTAL HEALTHCARE COST

-$13,146

healthcare-sector perspective; negative means the modelled programme saves money

REFERENCE-CASE NMB

$21,343

deterministic, at $100,000 per QALY — health valued at that threshold, not cash

DECISION UNCERTAINTY

1,500 of 1,500

modelled simulations cost-effective (100%); cost-saving in 1,499 of 1,500 simulations (>99%) · 95% interval $9,194 to $43,181 across 61 varied parameters

20

findings carry a standardised economic estimate

2 high-confidence, 11 moderate-confidence, 8 low-confidence, 22 ungraded. 23 of the 43 findings shown are reported without a dollar figure rather than being assigned one the model cannot support.

- LDLR familial hypercholesterolemia
- APOB familial hypercholesterolemia
- APOE ε4 carrier
- PTPN22 R620W carrier
- CHEK2 I157T carrier
- Factor V Leiden / Prothrombin...
- HLA-DQ2/DQ8 carrier
- predicted-pathogenic 12:111803962
- predicted-pathogenic 4:46312575
- HBB carrier
- SERPINA1 carrier
- OPRM1 OPRM1 A118G (mu-opioid...
- LRRK2 G2019S carrier
- G6PD deficiency
- DRD2/ANKK1 DRD2 / ANKK1 Taq1A
- NAT2 — NAT2 slow acetylator —...
- PON1 — PON1 Q192R —...
- ATP7B variant
- Coeliac-permissive HLA haplotype...
- Vitamin D Synthesis Efficiency
- 2 actionable wellness variant(s)
- Avoid...
- OPRM1-guided analgesic dosing +...
- Dopamine-aware addiction...
- Aromatic amine exposure...
- Pesticide exposure avoidance +...
- Awareness for autoimmune symptom...
- Targeted supplementation
- Awareness for high-risk situations
- Serum copper/ceruloplasmin...
- Risk-aware screening schedule
- Partner testing for reproductive...
- Intensive cardiovascular risk...
- High-intensity statin + cascade...
- High-intensity statin therapy
- Negative DQ2/DQ8 rules out celiac
- tTG-IgA serology if symptomatic,...
- Prodromal PD monitoring +...
- Personalized AUD risk awareness +...
- OPRM1-guided opioid dosing +...
- CHRNA5-guided smoking cessation
- Trauma-informed care +...
- Reproductive genetics

What rises to the top

Ranked by expected net monetary benefit, each valued on its own.
Finding-level values are not additive; the pooled reference case is calculated separately.

LDLR FAMILIAL HYPERCHOLESTEROLEMIA

$9,442

no action recorded

HIGH

APOB FAMILIAL HYPERCHOLESTEROLEMIA

$6,658

no action recorded

HIGH

APOE E4 CARRIER

$3,514

no action recorded

MODERATE

PTPN22 R620W CARRIER

$2,509

no action recorded

MODERATE

How GenomeLens avoids counting the same benefit twice

GENOMELENS ECONOMICS

Your findings at a glance

SYNTHETIC INPUT

Findings that could change a decision

Every finding the model examined, the decision it bears on, the strength of the evidence, and what it is worth on its own. No dollar figure appears without the genomic result that produced it — the detail behind each figure is on pages 3 and 4.

RISK & PREVENTION 13 findings

LDLR familial hypercholesterolemia (heterozygous)	—	HIGH	$9,442
APOB familial hypercholesterolemia (heterozygous)	—	HIGH	$6,658
APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)	—	MODERATE	$3,514
PTPN22 R620W carrier — autoimmune screening awareness	—	MODERATE	$2,509
CHEK2 I157T carrier — moderate cancer risk awareness	—	MODERATE	$2,449
Factor V Leiden / Prothrombin carrier — situational anticoagulation	—	MODERATE	$2,273
HLA-DQ2/DQ8 carrier — celiac rule-out capability	—	MODERATE	$2,097
predicted-pathogenic 12:111803962	Confirm the predicted variant clinically…	MODERATE	$1,995
predicted-pathogenic 4:46312575	Confirm the predicted variant clinically…	MODERATE	$1,893
HBB carrier (reproductive)	—	MODERATE	$1,226
SERPINA1 carrier (reproductive)	—	MODERATE	$1,226
G6PD deficiency — oxidative drug avoidance	—	MODERATE	$463
Coeliac-permissive HLA haplotype (DQ2.5/DQ8) — serology decision	—	MODERATE	-$122

LOWER-CONFIDENCE & EXPLORATORY 8 findings

OPRM1 OPRM1 A118G (mu-opioid receptor) — OPRM1 A118G…	—	LOW	$699
LRRK2 G2019S carrier — Parkinson's early intervention	—	LOW	$613
DRD2/ANKK1 DRD2 / ANKK1 Taq1A — DRD2/ANKK1 Taq1A — reward-pathway…	—	LOW	$442
NAT2 — NAT2 slow acetylator — occupational cancer risk awareness	—	LOW	$90
PON1 — PON1 Q192R — organophosphate susceptibility	—	LOW	$90
ATP7B variant — Wilson's disease copper monitoring	—	LOW	-$9

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype findings with a standardised economic pathway.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

SYNTHETIC INPUT

Findings with weaker evidence, reported anyway

Lower confidence changes how a finding should be read, not whether it is shown. A negative expected NMB means the modelled cost of acting exceeds the modelled benefit — useful information, not an error. Evidence grade reflects the strength of the association, not the size of the value; provenance is on page 8.

LDLR familial hypercholesterolemia (heterozygous)

HIGH

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB **$9,442** medical cost averted **$9,136** health gain **0.008 QALY** cost to act **$500**

APOB familial hypercholesterolemia (heterozygous)

HIGH

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB **$6,658** medical cost averted **$6,578** health gain **0.006 QALY** cost to act **$500**

APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk)

MODERATE

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB **$3,514** medical cost averted **$3,869** health gain **0.001 QALY** cost to act **$500**

PTPN22 R620W carrier — autoimmune screening awareness

MODERATE

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB **$2,509** medical cost averted **$2,579** health gain **0.004 QALY** cost to act **$500**

CHEK2 I157T carrier — moderate cancer risk awareness

MODERATE

POTENTIAL DECISION

No standardised action pathway recorded.

KEY ASSUMPTION

Baseline event probability and the effect of acting are registered assumptions varied in the sensitivity analysis; neither is measured for this individual.

EXPECTED NMB **$2,449** medical cost averted **$2,579** health gain **0.004 QALY** cost to act **$500**

Reported without a standardised value: 2 actionable wellness variant(s), Avoid primaquine/dapsone/rasburicase..., OPRM1-guided analgesic dosing + AUD treatment..., Dopamine-aware addiction treatment +..., Aromatic amine exposure monitoring + bladder..., Pesticide exposure avoidance + cholinesterase..., Awareness for autoimmune symptom clusters, Targeted supplementation (vitamin..., Awareness for high-risk situations (surgery..., Serum copper/ceruloplasmin monitoring + early..., Risk-aware screening schedule, Partner testing for reproductive risk..., Intensive cardiovascular risk reduction..., High-intensity statin + cascade screening, High-intensity statin therapy,

Trauma-informed care + HPA-axis-targeted..., Reproductive genetics — 12 carrier.... No registry-backed economic pathway exists for these

SYNTHETIC INPUT

Taken together, what do these findings mean?

This is the canonical reference case. Findings bearing on the same condition are combined on the risk scale, each condition's cost of illness is charged exactly once, and trial efficacy is discounted to real-world adherence before any value is credited.

INCREMENTAL HEALTHCARE COST	INCREMENTAL QALYS	REFERENCE-CASE NMB	CLASSIFICATION
-$13,146 healthcare-sector perspective	**0.082** about 30 days of healthy life on average	**$21,343** at $100,000 per QALY	**dominant (more health, lower cost)** not defined — strategy dominates or is dominated

Condition pools — combined before value is credited

CONDITION	FINDINGS	BASELINE	RAW UNCORRECTED	POOLED	ADHERENCE	REALISED	COST AVERTED	QALYS	NMB
CAD	3	50.0%	98%	60%	50%	34%	$7,410	0.022	**$9,339**
Pathogenic	8	20.0%	290%	60%	65%	49%	$5,111	0.043	**$8,851**
Alzheimer	1	15.0%	30%	30%	35%	11%	$1,630	0.001	**$1,516**
IronOverload	3	20.0%	32%	17%	65%	12%	$478	0.010	**$1,148**
Depression	2	20.0%	18%	13%	50%	7%	$379	0.007	**$817**
Coeliac	1	20.0%	30%	30%	65%	20%	$283	0.0001	**-$29**

Raw is an uncorrected sum, not a modelled result. It adds the risk reductions of findings that share a condition, which is not how probabilities combine. Any value above 100% is the artifact this correction removes. The pooled figure beside it is the model's answer.

Two corrections, applied in order

The engine reports these as one combined reduction. They rest on different evidence and are separated here: overlap is a structural correction, adherence a behavioural one.

NAIVE AVOIDED COST		POOLED, AT TRIAL EFFICACY		REALISED BENEFIT
$39,330	−$13,361 condition pooling →	**$25,969**	−$10,678 real-world adherence →	**$15,291**
every finding counted independently		34% removed — one liability, one cost of illness		41% further removed — efficacy is not effectiveness

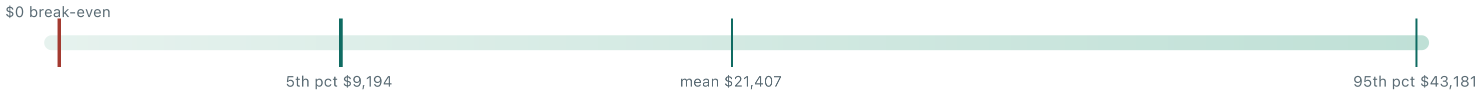
Adherence is charged to the benefit *and* to ongoing intervention cost, but not to the one-off test cost. 0.039 QALYs are lost to imperfect adherence relative to trial efficacy.

$21,343

How sure is this, and what rests on judgement?

Uncertainty has two layers: the spread of modelled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across 1,500 simulations



COST-EFFECTIVE

all 1,500 of 1,500 simulations

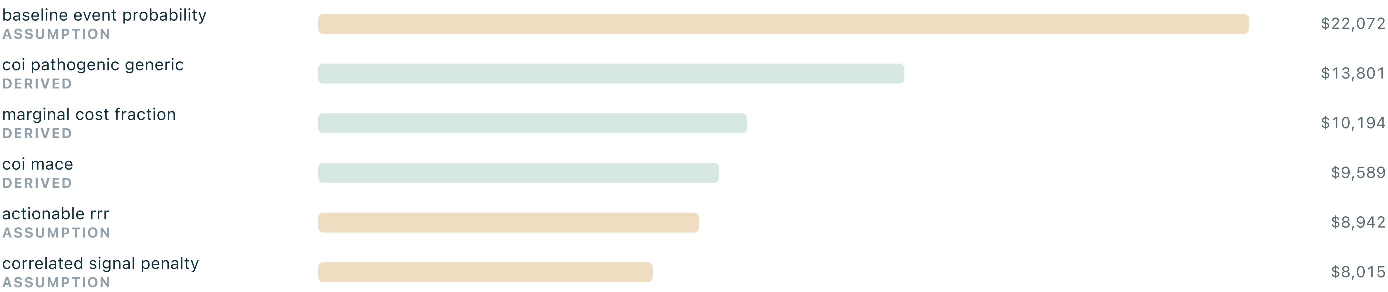
at $100,000 per QALY (100%). Reference-case NMB is $21,343; the mean across simulations is $21,407. Both are correct and answer slightly different questions.

COST-SAVING

1,499 of 1,500 simulations

the modelled programme costs less than usual care (>99%). Parameters without a published spread are held fixed; the interval below therefore understates true uncertainty rather than…

What drives the answer



Swing in net monetary benefit across each parameter's plausible range. Amber bars are declared assumptions rather than sourced values; they account for 47.5% of total swing.

EVIDENCE FOUNDATION

Registry sourced	**49 / 65 · 75.4%**
Whole model — PMID/DOI	**49.3%**
Whole model — attributed	**95.7%**
Whole model — judgement only	**4.3%**

Two different denominators — 65 registered parameters and 371 known model parameters. Reported separately, because collapsing them into one figure would create an inconsistency rather than remove one.

WHAT COULD CHANGE THE ANSWER

A meaningful share of this answer rests on judgement rather than published evidence. The direction is more trustworthy than the exact numbers.

0 ERRORS report-consistency validation — rendering permitted

4 warnings · 5 information notes, listed in full in the technical appendix

SYNTHETIC INPUT

Is whole-genome sequencing worth it?

Two different questions live on this page and they are never added together. One is a choice made *before* testing, averaged over a population. The other counts what sequencing actually found in this genome.

PROSPECTIVE — BEFORE TESTING

What is the expected value of choosing sequencing rather than an array?

Chance an array misses an actionable finding	**1 in 56**
Value if you are that person	**$25,189**
Expected value, averaged over everyone	**$491**
Extra cost over an array	**$200**
Net expected value	**$291**

OBSERVED — IN THIS GENOME

What did sequencing actually find that an array could not?

Sequencing-only findings present	**4**
Observed standalone contribution	**$6,340**

A count of findings this genome contains, not an expectation. For this input the population question is already settled.

Separation rule. The prospective figures describe the population this genome was drawn from; the observed figure describes the genome itself. They answer different decision questions, come from different inputs, and are never combined into one headline total.

Strategy frontier

STRATEGY	COST	QALYS	ICER VS PREVIOUS	STATUS
No testing	$0	20.000	—	efficient
Genotyping chip	$200	20.060	$3,333/QALY	efficient
Whole-genome sequencing	$600	20.120	$6,667/QALY	efficient
Targeted clinical panel	$900	20.090	—	dominated

Frontier values illustrate the testing choice for a population and are not this genome's reference case. This is an average over a population, and the distribution is extremely skewed: the great majority of people get nothing, and a small number get something that changes their care substantially. An expected value is the right…

SYNTHETIC INPUT

What was assumed, where it came from, what was checked

The first seven pages tell the story; this page is their audit trail. Full parameter tables, references and the advanced analyses follow in the technical appendix.

MODEL

Perspective	**healthcare sector**
Comparator	**usual care**
Outcomes	**cost, QALYs, NMB**
Time horizon	**10 years**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Pooling	**complement of products**
Adherence	**benefit and ongoing cost**
Uncertainty	**1,500-run PSA**

PROVENANCE

Registered parameters	**65**
Published	**8**
Derived	**41**
Declared assumptions	**16**
Registry sourced	**75.4%**
Model, resolvable	**49.3%**
Model, attributed	**95.7%**
Model, judgement only	**4.3%**

A parameter declared as an assumption may not cite a source — the registry refuses to load if one does.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labelled as such.
- Costs are not inflation-normalised to a single price year.
- Findings without a registry-backed pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Validation

PASS NMB identity — λ × ΔQALY − ΔCost reconciles to the reported total
PASS Condition cost charged once — no pool exceeds its cost of illness
PASS WGS basis separation — observed and prospective never combined
PASS Report validation gate — no errors; rendering permitted
PASS Dominance classification — sign of cost and effect agrees with the label
PASS Pooled risk cap — combined risk reduction stays within its bound
PASS Budget-impact consistency — peak burden is the maximum net spend

Audit trail. Every figure in this report is serialised in `economics-payload.json`, the canonical object these pages render from. The parameter registry, references, cross-path reconciliation and the advanced analyses follow in the technical appendix. Build GENOMELENS-BUILD: 7b610d5+dirty 2026-09-04T17:32:45-04:00. Synthetic input — no human genome was used.